Engineering Mechanics

Lecture-IV

Equilibrium in 2D

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Equilibrium

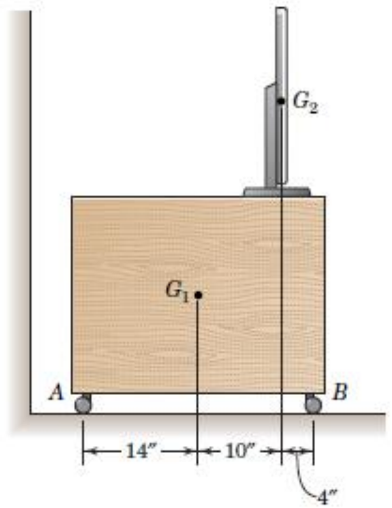
- When a body is in equilibrium, the resultant of *all* forces acting on it is zero.

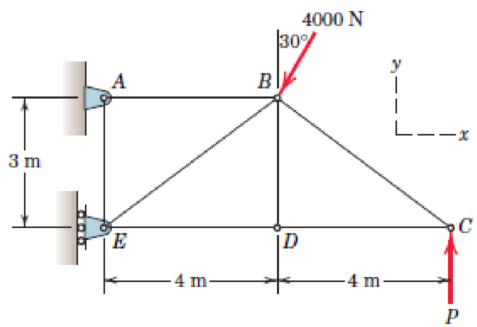
- Thus, the resultant force **R** and the resultant couple **M** are both zero, and we have the equilibrium equations

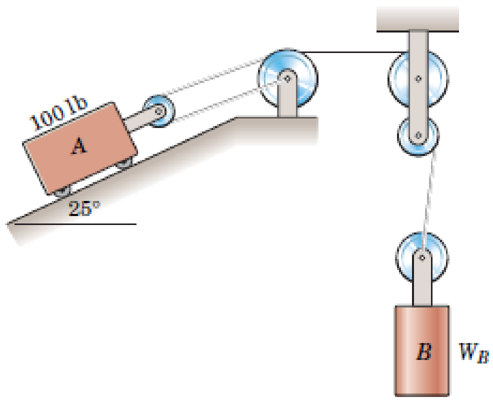
$$\boxed{\mathbf{R} = \Sigma\mathbf{F} = 0}$$

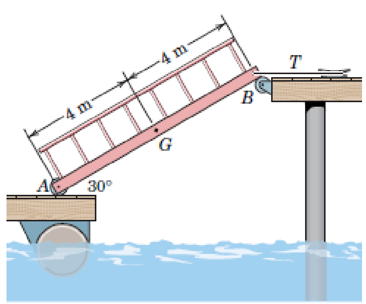
$$\boxed{\mathbf{M} = \Sigma\mathbf{M} = 0}$$

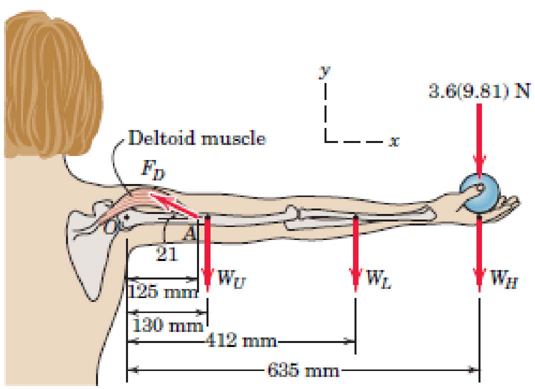
EQUILIBRIUM IN TWO DIMENSIONS

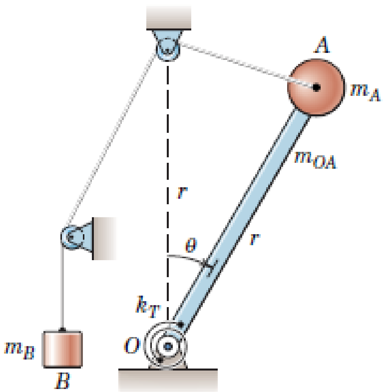












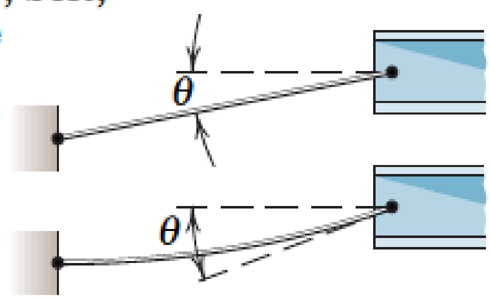
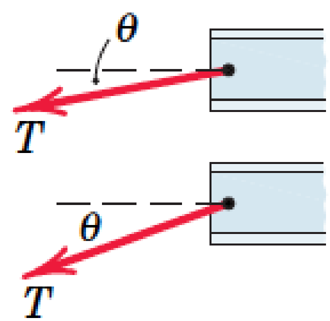
Free-body diagram

- The free-body diagram is the most important single step in the solution of problems in mechanics.

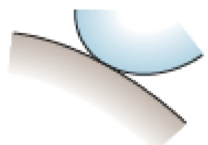
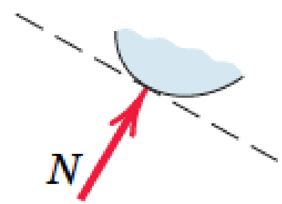
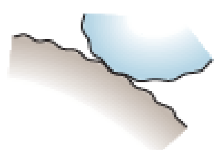
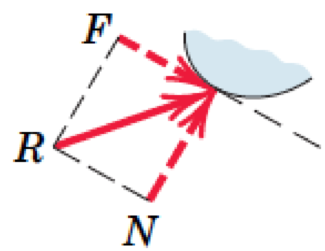
CONSTRUCTION OF FREE-BODY DIAGRAMS

- ***Step 1.*** Decide which system to isolate.
- ***Step 2.*** Next isolate the chosen system by drawing a diagram which represents its *complete external boundary*.
- ***Step 3***. Identify and represent all forces which act on the isolated system as applied by the removed contacting and attracting bodies
- ***Step 4.*** Show the choice of coordinate axes directly on the diagram. Pertinent dimensions may also be represented for convenience.

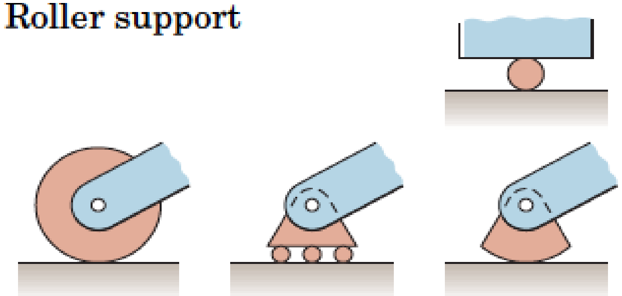
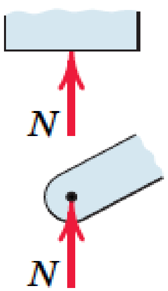
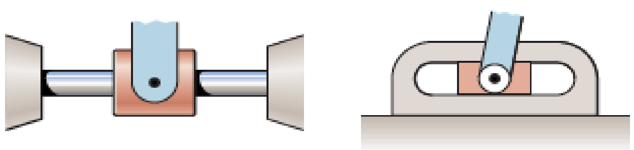
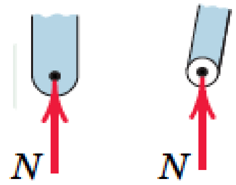
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
1. Flexible cable, belt, chain, or rope Weight of cable negligible Weight of cable not negligible  	 Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.

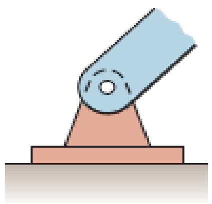
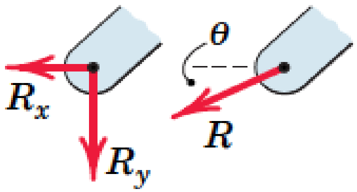
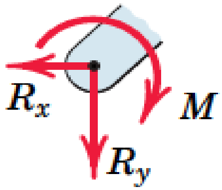
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
2. Smooth surfaces 	 Contact force is compressive and is normal to the surface.
3. Rough surfaces 	 Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R.

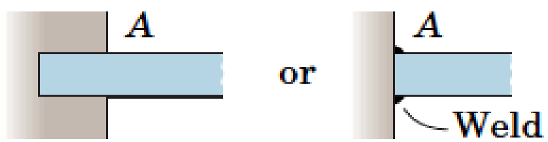
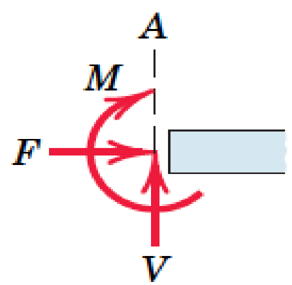
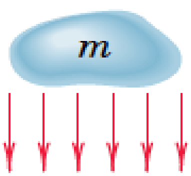
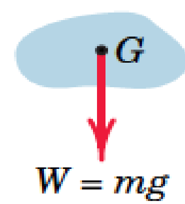
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
4. Roller support 	 Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.
5. Freely sliding guide 	 Collar or slider free to move along smooth guides; can support force normal to guide only.

Action of force in 2D

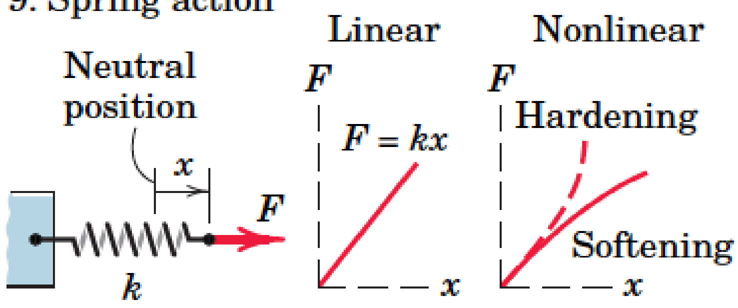
MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS (*cont.*)	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
6. Pin connection 	Pin free to turn  Pin not free to turn  A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis. We may either show two components R_x and R_y or a magnitude R and direction θ. A pin not free to turn also supports a couple M.

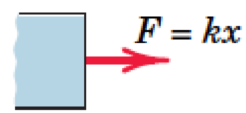
Action of force in 2D

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS	
Type of Contact and Force Origin	**Action on Body to Be Isolated**
7. Built-in or fixed support 	 A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation.
8. Gravitational attraction 	 The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center of gravity G.

Action of force in 2D

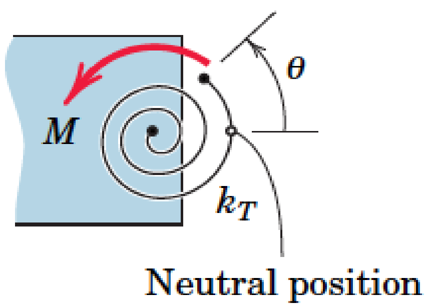
9. Spring action

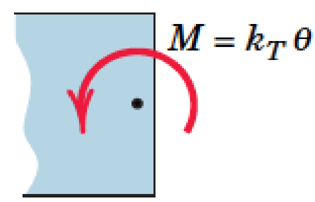




Spring force is tensile if the spring is stretched and compressive if compressed. For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

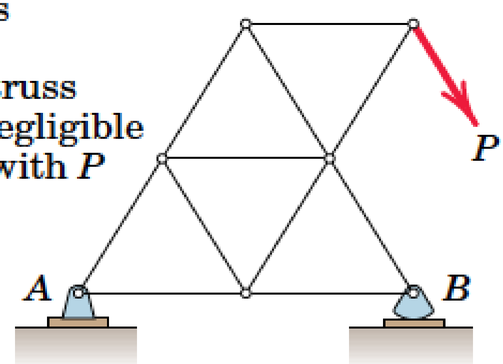
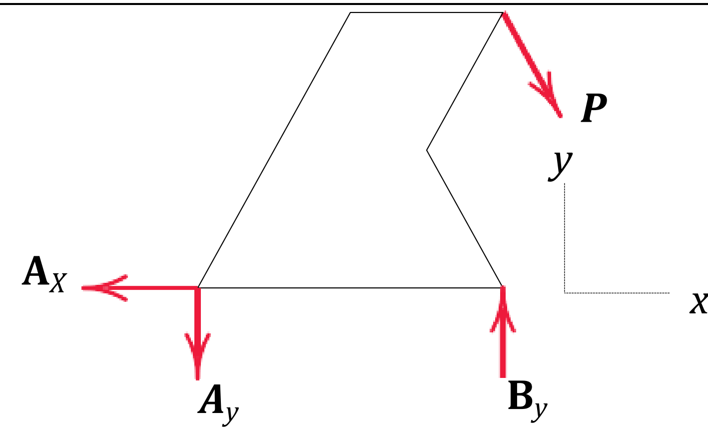
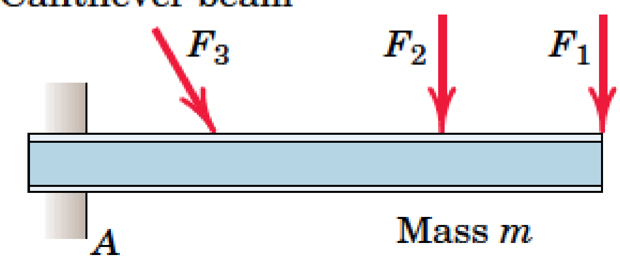
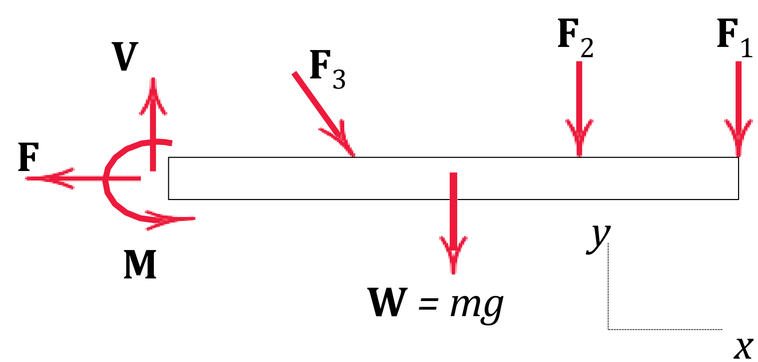
10. Torsional spring action



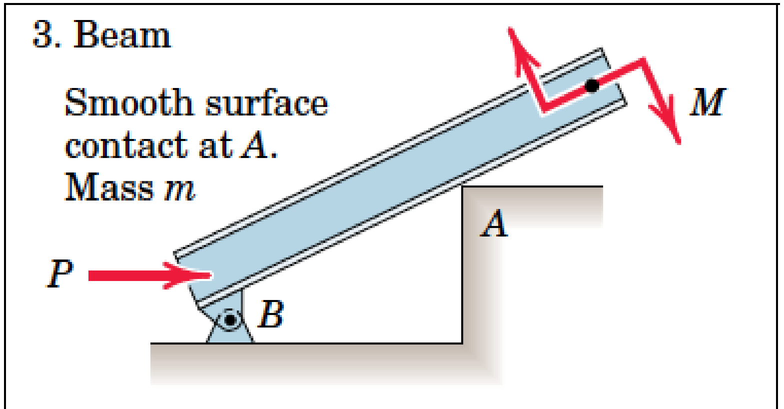


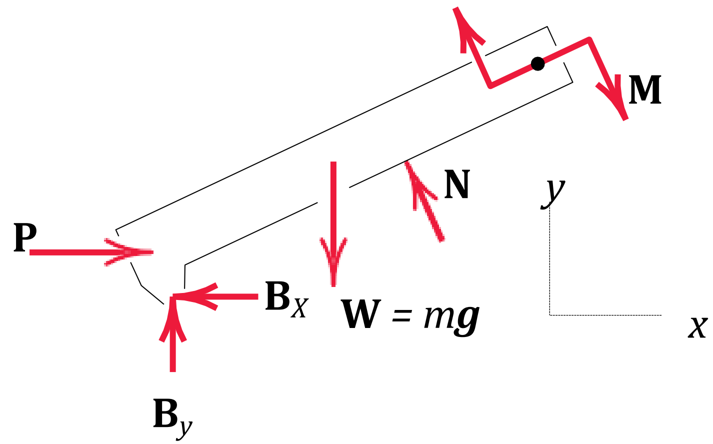
For a linear torsional spring, the applied moment M is proportional to the angular deflection θ from the neutral position. The stiffness k_T is the moment required to deform the spring one radian.

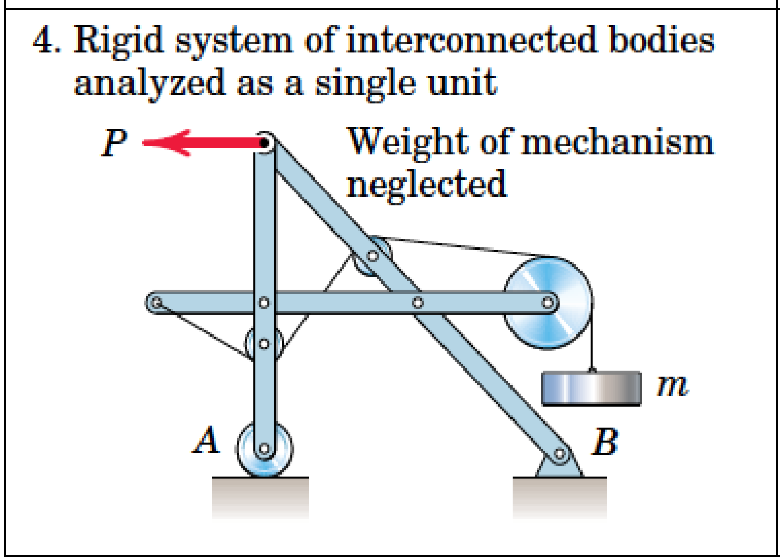
Freebody Diagram in 2D

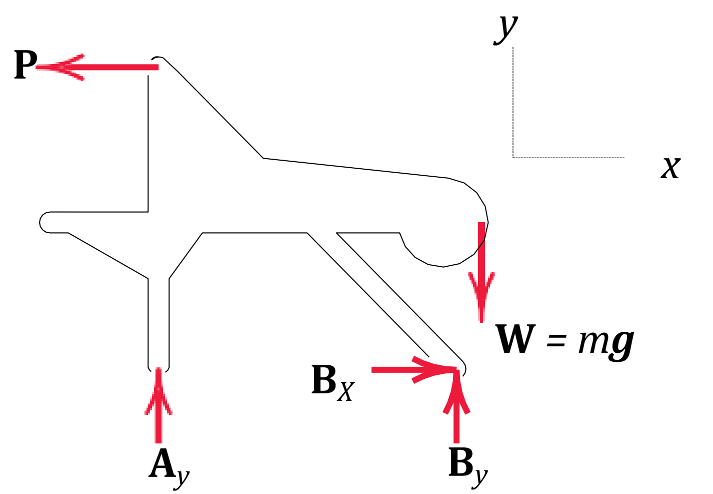
SAMPLE FREE-BODY DIAGRAMS	
Mechanical System	**Free-Body Diagram of Isolated Body**
1. Plane truss Weight of truss assumed negligible compared with P  	
2. Cantilever beam  	

Freebody Diagram in 2D







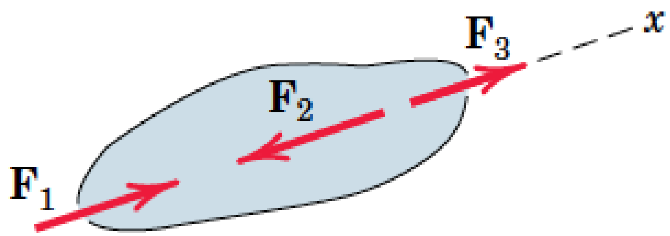
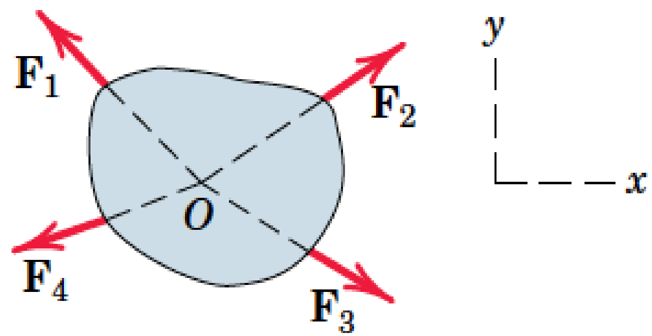


EQUILIBRIUM CONDITIONS (2D)

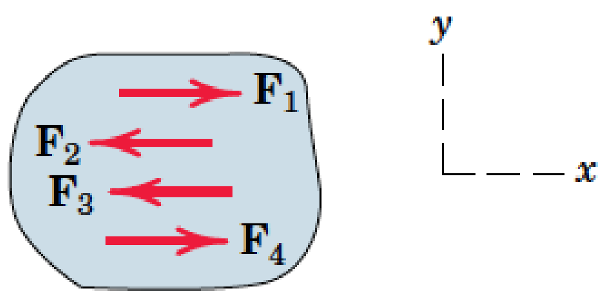
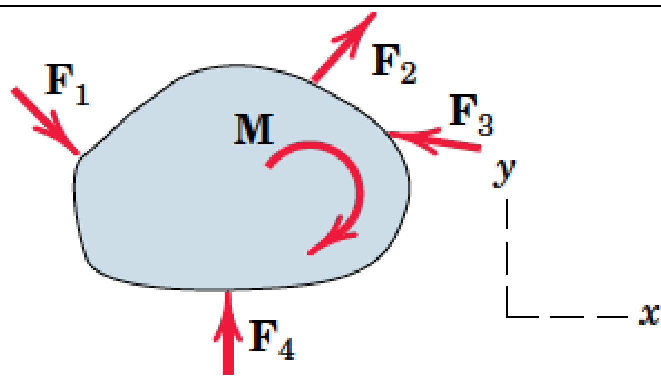
- **Equilibrium**: the condition in which the resultant of all forces and moments acting on a body is zero
- which in two dimensions may be written in scalar form as

$$\boxed{\begin{gathered}\Sigma F_x = 0\\ \Sigma F_y = 0\\ \Sigma M_o = 0\end{gathered}}$$

Categories of equilibrium

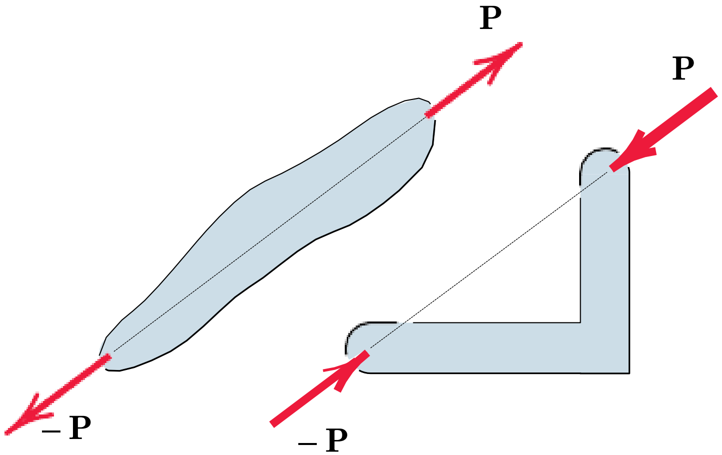
CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
1. Collinear		$\Sigma F_x = 0$
2. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$

Categories of equilibrium

CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
3. Parallel		$\Sigma F_x = 0 \quad \Sigma M_z = 0$
4. General		$\Sigma F_x = 0 \quad \Sigma M_z = 0$ $\Sigma F_y = 0$

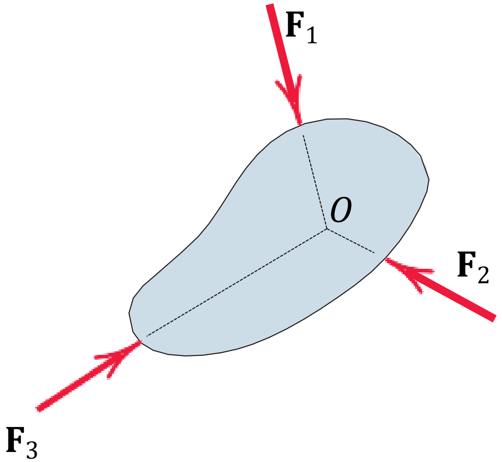
Two- Force Members

- A *two-force member* to be in equilibrium, the forces must be *equal, opposite,* and *collinear*

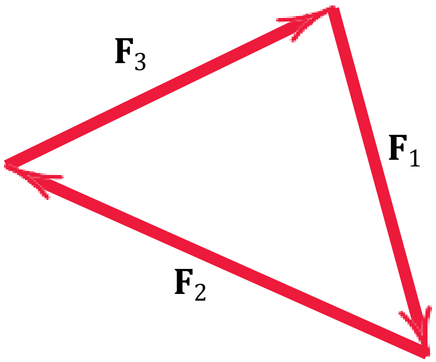


Three-Force Members

- We see that equilibrium requires the lines of action of the three forces to be *concurrent*.
- If they were not concurrent, then one of the forces would exert a resultant moment about the point of intersection of the other two



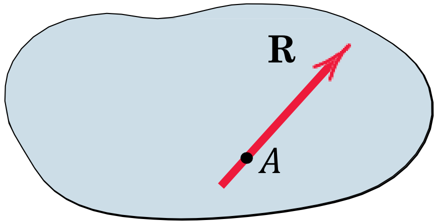
$$\boxed{\Sigma \mathbf{F} = 0}$$



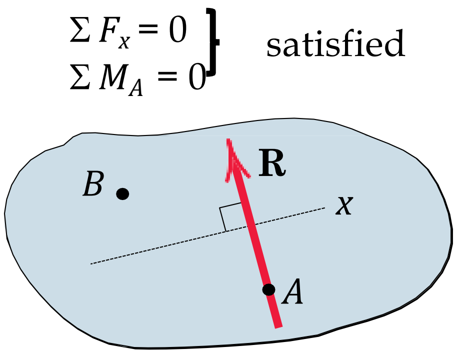
2nd Alternative Equilibrium Equations

1. $\Sigma M_A = 0$; Resultant may still exist, cannot be a couple, but must be a force R passing through A.
2. $\Sigma F_x = 0$; Resultant may still exist, not only must pass through A, but also must be perpendicular to the x-direction as shown.
3. $\Sigma M_B = 0$; Resultant can not exist if B is any point such that the line AB is not perpendicular to the x-direction

These form second set of alternative Equilibrium Equations.



(a)



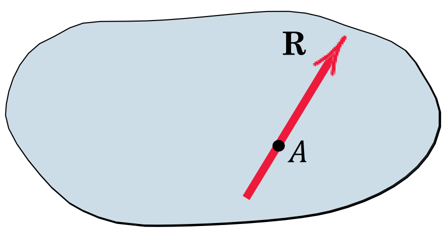
(b)

3^{rd} Alternative Equilibrium Equations

1. $\Sigma\, M_A = 0$; Resultant may still exist
2. $\Sigma\, M_B = 0$; Resultant may still exist
3. $\Sigma\, M_C = 0$; Resultant can not exist if A, B, and C are any three points not on the same straight line.

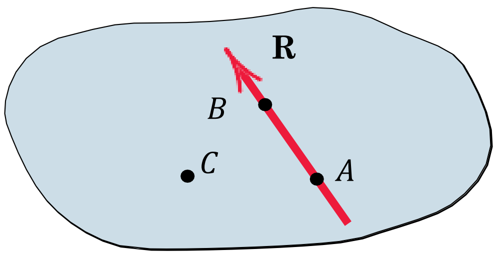
These form third set of alternative Equilibrium Equations.

$\Sigma\, M_A = 0$ satisfied



(c)

$$\left.\begin{aligned}\Sigma\, M_A &= 0\\ \Sigma\, M_B &= 0\end{aligned}\right\}\ \text{satisfied}$$



(d)

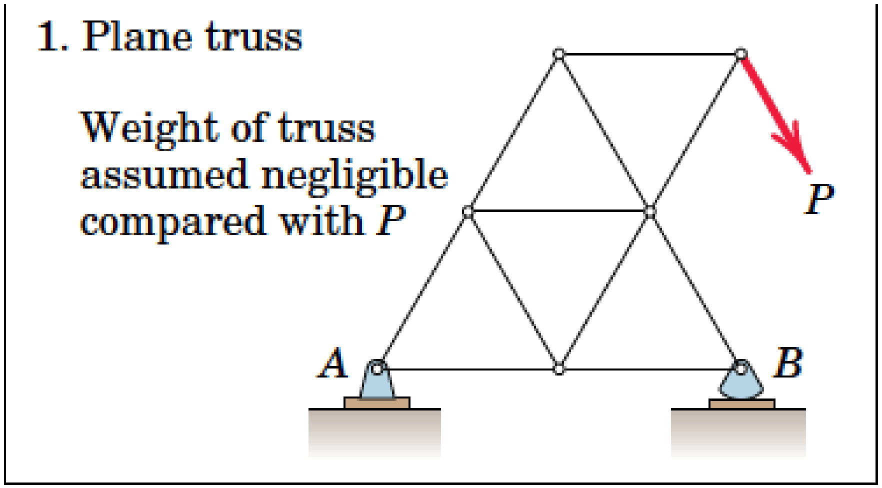
Constraints and Statically Determinacy

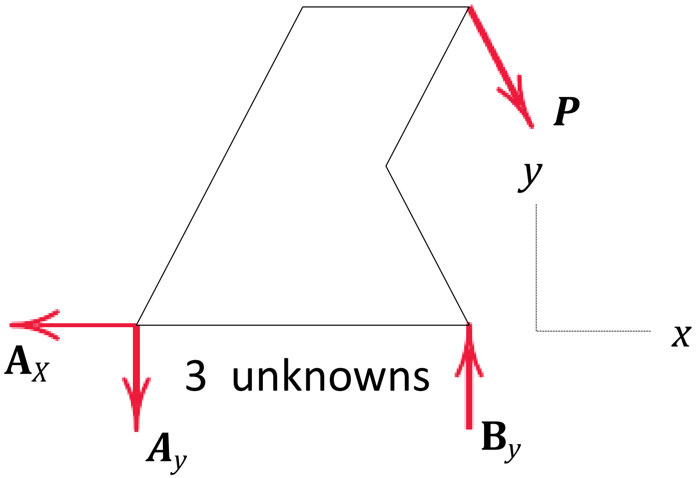
- The equilibrium equations are both necessary and sufficient conditions to establish the equilibrium of a body.

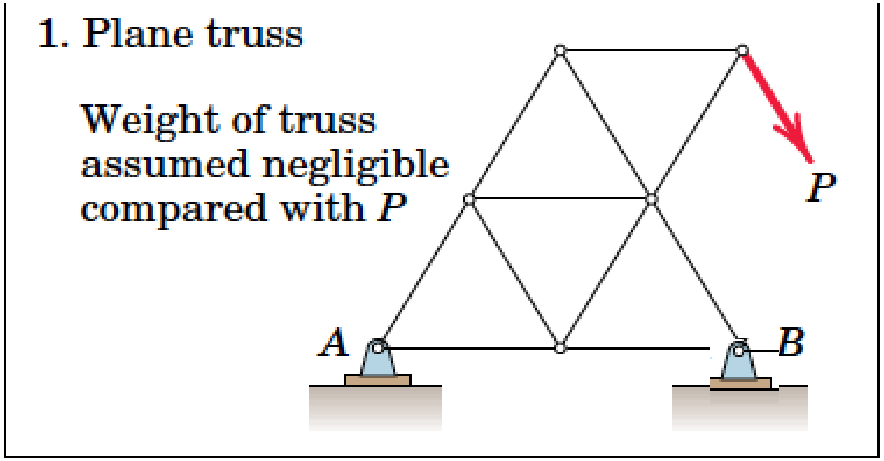
- Whether the equations are adequate to determine all the unknowns depends on the characteristics of the constraints provided by its supports.

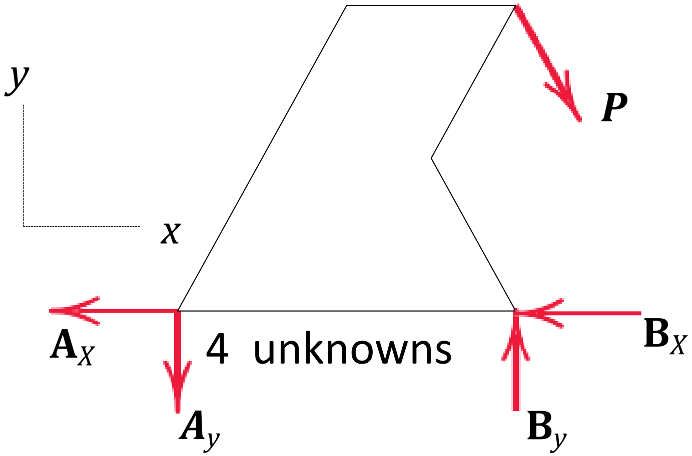
- ***Constraints*** mean the restriction of movement.

Number of unknowns









Statically indeterminate

- A rigid body, or rigid combination of elements treated as a single body, which possesses more external supports or constraints than are necessary to maintain an equilibrium position is called *statically indeterminate*.

- Supports which can be removed without destroying the equilibrium condition are said to be *redundant*.

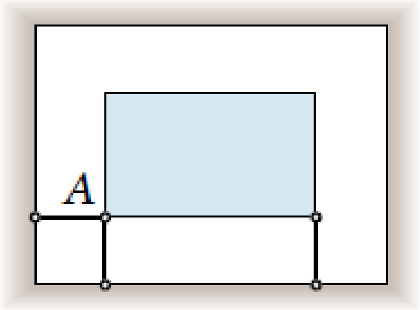
- *Degree of statical indeterminacy:* the total number of unknown external forces, minus the number of available independent equations of equilibrium.

Statically Determinate

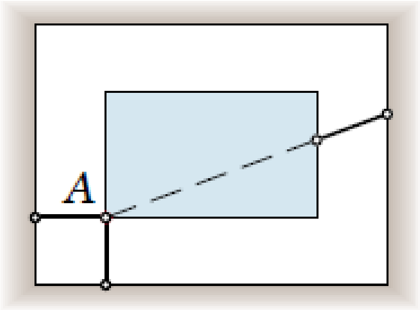
- Bodies which are supported by the minimum number of constraints necessary to ensure an equilibrium configuration are called *statically determinate,*

- for such bodies the equilibrium equations are sufficient to determine the unknown

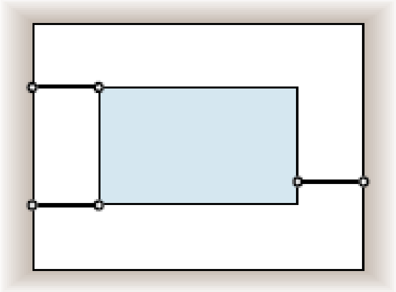
Adequacy of Constraints



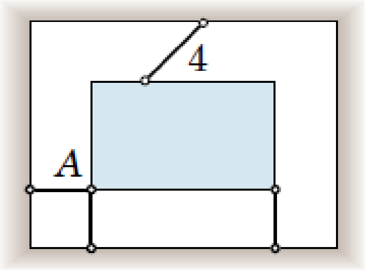
(*a*) Complete fixity
Adequate constraints



(*b*) Incomplete fixity
Partial constraints



(*c*) Incomplete fixity
Partial constraints



(*d*) Excessive fixity
Redundant constraint

APPROACH TO SOLVING PROBLEMS

1. Identify clearly the quantities which are known and unknown

2. Make an unambiguous choice of the body to be isolated and draw its complete free-body diagram

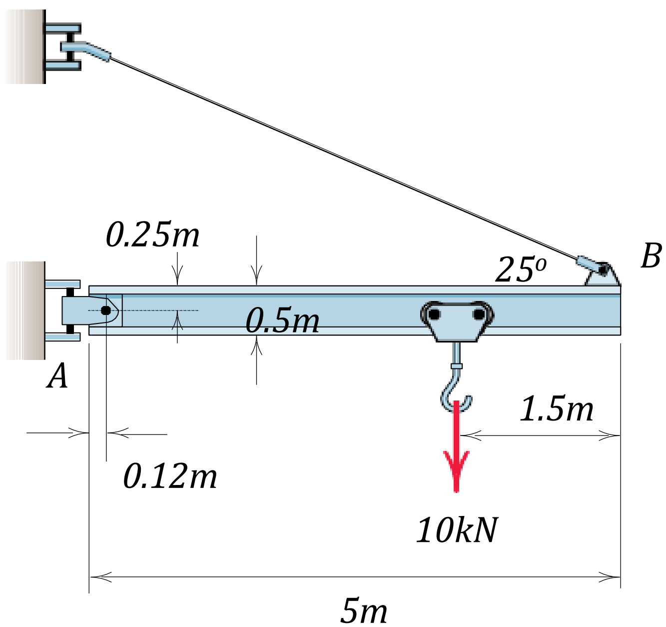
3. Choose a convenient set of reference axes, always using righthanded axes when vector cross products are employed.

4. Identify and state the applicable force and moment principles or equations which govern the equilibrium conditions of the problem.

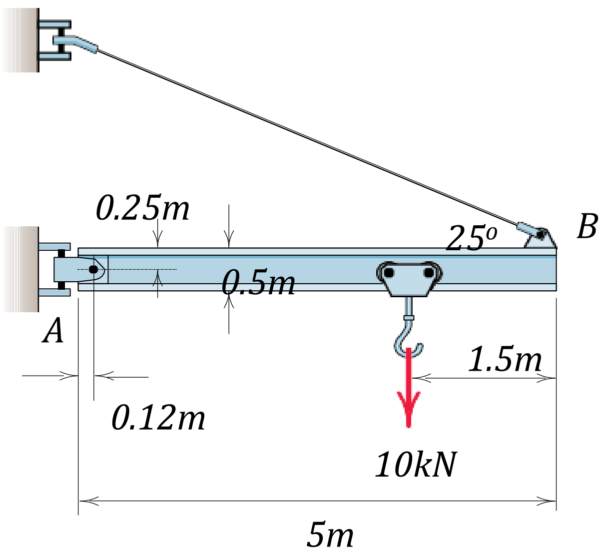
5. Match the number of independent equations with the number of unknowns in each problem.

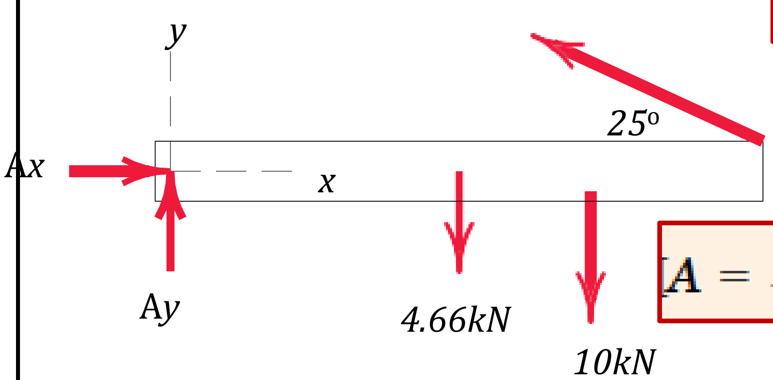
6. Carry out the solution and check the results

- **Example 1:** Determine the magnitude T of the tension in the supporting cable and the magnitude of the force on the pin at A for the jib crane shown. The beam AB is a standard 0.5-m I-beam with a mass of 95 kg per meter of length.



Solution:





$$[\Sigma M_A = 0]$$

$$(T\cos 25°)0.25 + (T\sin 25°)(5 - 0.12)$$
$$- 10(5 - 1.5 - 0.12) - 4.66(2.5 - 0.12) = 0$$

$$T = 19.61 \text{ kN}$$

$$[\Sigma F_x = 0]$$

$$A_x - 19.61\cos 25° = 0$$

$$A_x = 17.77 \text{ kN}$$

$$[\Sigma F_y = 0]$$

$$A_y + 19.61\sin 25° - 4.66 - 10 = 0$$

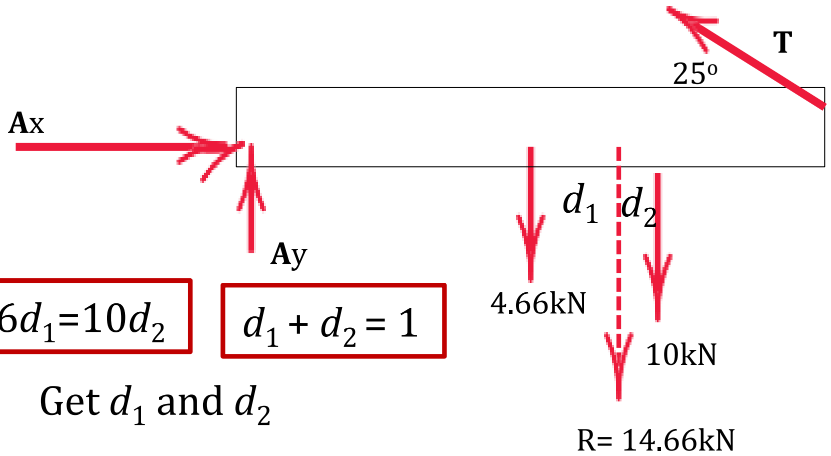
$$A_y = 6.37 \text{ kN}$$

$$[A = \sqrt{A_x{}^2 + A_y{}^2}]$$

$$A = \sqrt{(17.77)^2 + (6.37)^2} = 18.88 \text{ kN}$$

Graphical Approach

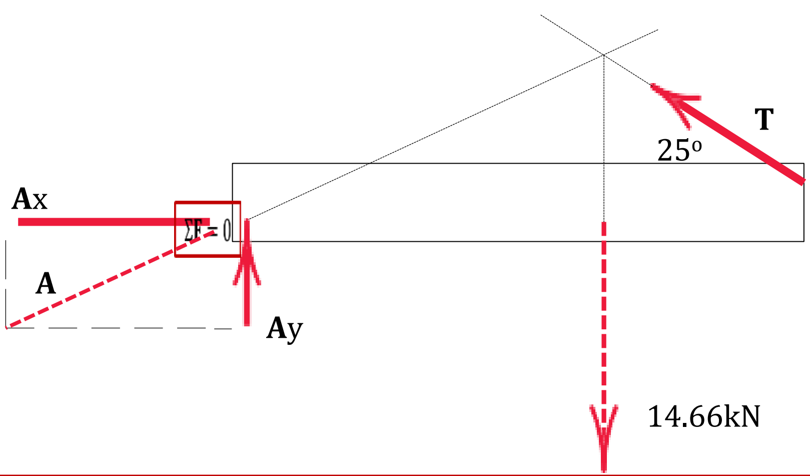
Step 1: Find resultant of 4.66 kN and 10kN



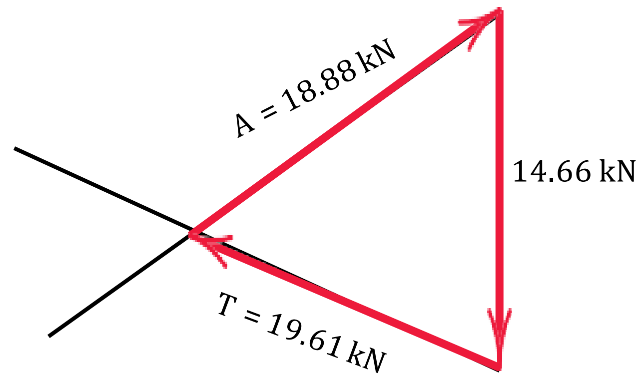
$\boxed{4.66d_1=10d_2}$ $\boxed{d_1 + d_2 = 1}$

Get d_1 and d_2

Step 2: Find direction of A



Step 3: Find magnitude of T and F



$\boxed{\Sigma \mathbf{F} = 0}$

THANK YOU